Topic 12: Cellular Concepts

CSE 4405: Data and Telecommunications

Department of CSE

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Learning Objectives

1. State the three system-level ideas behind the cellular concept
2. Explain why a regular cell shape is assumed, and why the hexagon in particular
3. Derive $N = i^2 + ij + j^2$ and $Q = \sqrt{3N}$ from hexagonal lattice geometry
4. Use the handoff threshold, dwell time, and prioritization schemes to reason about mobility
5. Compute first-order SIR, invert it to find a minimum cluster size, and state what it ignores
6. Distinguish boundary coverage from area coverage under shadowing
7. Apply trunking vocabulary: offered traffic, Erlang, grade of service, trunking efficiency
8. Compare FCA, DCA, and hybrid assignment, and the expansion techniques of splitting and sectoring

The Cellular Concept

Why it mattered

The cellular concept was the breakthrough that solved spectral congestion and user capacity. It delivered very high capacity in a limited spectrum allocation **without any major technological change**: it is an architectural idea, not a new radio.

1. Divide

Replace a single high-power transmitter with many low-power transmitters, each covering a small area.

2. Separate

Neighboring cells get *different* channel groups, so adjacent coverage does not collide.

3. Reuse

The same channel set is used again at a geographically distant location.

The Design Problem

The goal

Provide good coverage *and* good service in a high user-density area. Both words matter: coverage alone is not service if every channel is busy.

When may we reuse?

Only once the total interference from all users in the co-channel cells has suffered **sufficient attenuation**.

What "sufficient" depends on

- **Geographical separation**: path loss grows with distance
- **Shadowing**: terrain and buildings add or remove attenuation unpredictably

Read this as a budget

Reuse is not free distance; it is bought with attenuation. Everything later in this deck is about spending that budget well.

Cell Footprint

The **cell footprint** is the *actual* radio coverage of a cell, as opposed to the shape drawn on a planning map.

Irregular is fine at first

In an initial system design, irregular cell structure and irregular transmitter placement may be perfectly acceptable. Coverage is the only goal and there are few cells to conflict.

Until traffic grows

As new cells and channels are added, irregularity leads to an **inability to reuse frequencies**, because there is no systematic way to certify that two cells are far enough apart. Co-channel interference is the symptom.

So we assume a regular shape

For *systematic* cell planning, a regular footprint is assumed. This is a bookkeeping decision, not a claim about radio physics.

Why Not Circles, and Why Hexagons

The contour should be circular

An omnidirectional antenna radiates equally in all directions, so the natural equal-power contour is a circle.

But circles are impractical

Circles do not tile. Pack them and you get **ambiguous areas**: regions with *no* coverage between them, or regions with *multiple* coverage where they overlap. A planner cannot assign a channel to an ambiguous area.

Only three regular polygons tile.

Area covered at the same reach R:

Triangle	$1.30\,R^2$
Square	$2.00\,R^2$
Hexagon	$\mathbf{2.60\,R^2}$
(circle)	$3.14\,R^2$

The economic argument

For the same transmitter reach, the hexagon covers the most ground: about 83% of the circle it sits inside. Most area per site means the **fewest base stations** per city. That is why hexagons won.

Reference Figure: The Hexagonal Layout

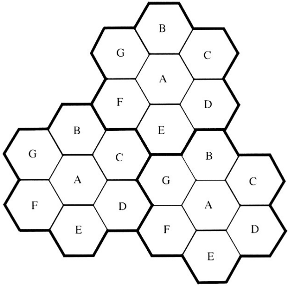


Figure 3.1 Illustration of the cellular frequency reuse concept. Cells with the same letter use the same set of frequencies. A cell cluster is outlined in bold and replicated over the coverage area. In this example, the cluster size, *N*, is equal to seven, and the frequency reuse factor is 1/7 since each cell contains one-seventh of the total number of available channels.

Rappaport 2e Fig. 3.1: a conventional cellular layout as a uniform grid of regular hexagons; same-letter cells share a channel set.

Frequency Reuse: $S = kN$

A system has S duplex channels. They are divided among N cells so that each cell uses a **unique and disjoint** set of k channels:

$S = kN$ hence $k = S/N$

What "disjoint" buys

Because the N sets do not overlap, no two cells inside one cluster can interfere on the same frequency. Interference is pushed outside the cluster by construction.

The whole tradeoff in one equation

S is fixed by your licence. So raising N to push co-channel cells apart *necessarily* shrinks k. You cannot improve interference and capacity at the same time by choosing N.

Terminology

Term	Definition
Cluster size N	The N cells which collectively use the complete set of available frequencies.
Co-channel cell	A cell using the same set of frequencies as the target cell.
Interference tier	A set of co-channel cells at the *same* distance from the reference cell.
First tier	The set of *closest* co-channel cells. It always contains exactly **six** cells.

The word “tier” is doing work

Interference arrives in rings, not as a blur. The first ring dominates because power falls off steeply with distance, which is why almost all planning arithmetic stops after the first tier.

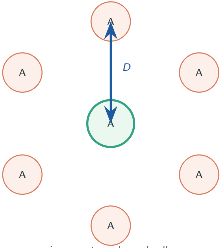
Why the First Tier Always Has Exactly Six Cells

Why six, always

The co-channel cells form their *own* triangular lattice. Every point of a triangular lattice has exactly six nearest neighbors, spaced 60° apart. Changing N changes *how far away* they sit, never *how many* there are.

Where this shows up

This is why $i_0 = 6$ appears later in the SIR formula. It is not a modelling convenience; it is lattice geometry.



six nearest co-channel cells,
60° apart, for every N

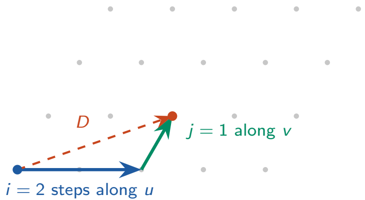
Hexagonal Coordinates: Two Axes at 60°

The trick

Do not use x and y. Use two axes u and v pointing at two adjacent cell centers, 60° apart. Both have length $d = \sqrt{3}R$.

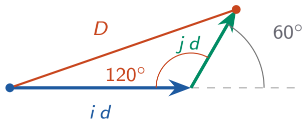
Why it pays off

With these co-ordinates, the center of *every* cell falls on a point specified by a pair of **integers** (i, j). Messy plane geometry becomes integer bookkeeping.



$(i, j) = (2, 1) \Rightarrow N = 7$

The Distance Between Two Cell Centers



Mind the angle

You *turn* by 60°, but the cosine rule needs the angle *inside* the triangle, which is the supplement $180^\circ - 60^\circ = 120^\circ$.

$$D^2 = (id)^2 + (jd)^2 - 2(id)(jd)\cos 120^\circ$$
$$= d^2\left(i^2 + j^2 + ij\right) \quad \text{since } \cos 120^\circ = -\tfrac{1}{2}$$

With $d = \sqrt{3}R$: $\quad D = \sqrt{i^2 + j^2 + ij} \cdot \sqrt{3}R$

Where the cross term is born

The $+ij$ is the $-2ab\cos C$ term, and it lands **positive** only because $\cos 120^\circ$ is negative. Run the same argument on a **square** grid, where you turn 90°: $\cos 90^\circ = 0$, the cross term vanishes, and you get plain Pythagoras. The entire ij term exists because the lattice is hexagonal.

Cluster Size Must Satisfy $N = i^2 + ij + j^2$

Cluster size: $N = i^2 + ij + j^2$, i, j non-negative integers.

Same expression, second job

The quantity under the square root in D *is* the cluster size. Both lattices are triangular, and their densities compare as the square of spacing, so $(D/d)^2 = N$ cells fall inside each co-channel cell's territory.

Allowed values are not arbitrary

Because i, j are integers, only certain N can occur:

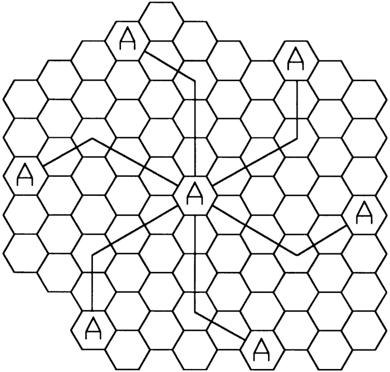
$1, 3, 4, 7, 9, 12, 13, 16, 19, 21, 25, \ldots$

Never $2, 5, 6, 8, 10, 11, 14, 15$.

The exact rule, if you want one

N is achievable exactly when every prime factor of N that leaves remainder 2 on division by 3 appears to an *even* power. That is why 2 and 5 are impossible but $4 = 2^2$ and $25 = 5^2$ are fine.

Reference Figure: Locating Co-Channel Cells



Rappaport 2e Fig. 3.2: move i cells along one hexagonal chain, turn 60°, then move j cells to reach a nearest co-channel cell.

Worked Example: the $N = 19$ Pattern

Question

A design uses shift parameters $i = 3$, $j = 2$. What cluster size results, and how far apart are co-channel cells?

Cluster size

$$N = i^2 + ij + j^2$$
$$= 9 + 6 + 4 = \mathbf{19}$$

Reuse geometry

$$D = \sqrt{19} \cdot \sqrt{3}R$$
$$Q = D/R = \sqrt{57} \approx \mathbf{7.55}$$

Sanity check

19 is on the allowed list because it was *built* from integers i, j. Any (i, j) you pick lands on a legal cluster size automatically; you can never accidentally construct $N = 5$.

Co-Channel Reuse Ratio Q

Reuse ratio: $$Q = \frac{D}{R} = \sqrt{3N}$$

D: distance between nearest co-channel cell centers. R: cell radius.

What Q measures

How far co-channel cells sit *relative to cell size*. It is a shape parameter, not a distance: doubling every cell leaves Q unchanged.

Quick values

$N = 3 \Rightarrow Q = 3$
$N = 4 \Rightarrow Q \approx 3.46$
$N = 7 \Rightarrow Q = \sqrt{21} \approx 4.58$
$N = 12 \Rightarrow Q = 6$

Keep the two ratios apart

$D/d = \sqrt{N}$ but $D/R = \sqrt{3N}$. The extra $\sqrt{3}$ is nothing but the hexagon's center-to-center spacing $d = \sqrt{3}R$, which comes from the apothem $\frac{\sqrt{3}}{2}R$ of an equilateral triangle.

Handover: What It Is

Handover / handoff

Transfer of an ongoing call, occurring when a mobile moves into a different cell during an existing call, or when it moves from one cellular system into another.

Three requirements

It must be **user transparent**, **successful**, and **not too frequent**. Failing any one of the three is a service defect.

More than finding a tower

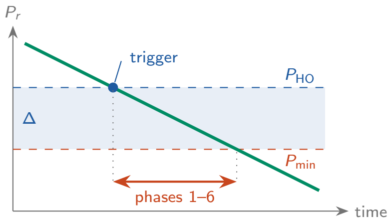
Handover not only identifies a new base station: it also requires that the voice and control signals be *allocated to channels* associated with the new BS. If that cell has no free channel, the handover fails even with perfect signal.

How a Handoff Actually Happens

The sequence

0 Measure serving *and* neighbours continuously; reports are **averaged**, never raw
1 Averaged level crosses P_{HO} and persists $\Rightarrow$ trigger
2 BSC/MSC chooses the target cell
3 Target reserves a channel $\leftarrow$ can fail
4 Handover command sent on the *old* link
5 Mobile retunes and synchronises $\leftarrow$ the gap
6 MSC switches the voice path over
7 Old channel released

BSC = Base Station Controller, owns the radio resources of many cells.
MSC = Mobile Switching Centre, the network's telephone switch.



This is what Δ buys

The signal keeps falling throughout phases 1–6, so the procedure must finish before the trace reaches $P_{\min}$. Trigger at $P_{\min}$ itself and you would finish well below it. Soft handoff *overlaps* phases 3–6 instead of running them in series, which is why it has no gap and almost no deadline.

The Handoff Threshold

P_{min}: the **minimum usable signal** for acceptable voice quality at the receiver.

P_{HO}: a *slightly stronger* level at which handover is initiated.

$$\Delta = P_{\text{HO}} - P_{\text{min}}, \quad \text{so } P_{\text{HO}} = P_{\text{min}} + \Delta$$

Reading dBm

dBm is absolute power against 1 mW: $10\log_{10}(P/1\,\mathrm{mW})$. Handset signals sit far below 1 mW, so the numbers are always negative (-100 dBm $=$ 0.1 pW). **Less negative means stronger**: $-85 > -92 > -100$.

dB alone is a *ratio*. Subtracting two dBm cancels the reference, which is why Δ is measured in dB, not dBm.

Why the gap exists at all

Handover takes time: measure, decide, signal, allocate, switch. If you waited until the signal was merely *usable*, there would be no time left to act. Δ buys the network that time, paid for in signal margin.

Worked Example: Up to the Trigger

BS-A at road-km 0, BS-B at km 4 (midpoint km 2.0). Car at 60 km/h. Suburban, $n = 3.5$, $P(d) = -78 - 35\log_{10}(d/1.5)$ dBm. $P_{\text{min}} = -100$, $\Delta = 8 \Rightarrow P_{\text{HO}} = -92$, hysteresis $H = 3$ dB, time-to-trigger (TTT) 2 s.

Time	Position	A (dBm)	B (dBm)	What happens
0 s	km 1.5	−78.0	−85.8	Phase 0: already measuring six neighbours every 480 ms
30 s	km 2.0	−82.4	−82.4	Exactly equal, still no handoff: B must beat A by the hysteresis H
42 s	km 2.2	−83.8	−80.8	$B - A = 3.0$ dB, condition met, TTT starts
44 s	km 2.23	−84.0	−80.6	Held the full 2 s $\Rightarrow$ **trigger** (phase 1)

Note what did not trigger it

A was at −83.8 dBm, a full 8 dB *above* $P_{\text{HO}} = -92$. The **better-cell** criterion fired; the absolute threshold never came into it.

Choosing Δ: Both Directions Hurt

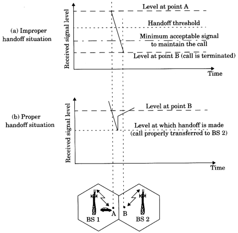


Figure 3.3 Illustration of a handoff scenario at cell boundary.

Rappaport 2e Fig. 3.3: improper (top) and proper (bottom) handoff situations at a cell boundary.

Δ too small

- Insufficient time to complete handoff before the call is lost
- More dropped calls

Δ too large

- Too many handoffs, many unnecessary
- Burden on the MSC

The reading

The network cannot see *where* you are, only your received power, so it cannot by itself tell a genuine departure from a passing truck. Δ therefore trades *reliability* against *signalling load*, and no single value is right everywhere.

Δ Too Small: the Corner Effect

Urban **microcell**, car at 50 km/h. With $P_{\min} = -100$ dBm and $\Delta = 3$ dB, $P_{\text{HO}} = -100 + 3 = -97$ dBm. Cruising with line of sight to BS-A at -88 dBm.

The car turns a corner

Line of sight to BS-A is lost behind the building on the corner. The signal falls ≈ 25 dB in about 4 m, roughly 0.3 s.

$-88 \to -113$ dBm: straight past P_{HO} *and* past $P_{\min}$, in one movement.

No value of Δ saves this

Decay rate $\approx 25/0.3 \approx 80$ dB/s. To cover the ≈ 0.8 s phase 2–7 sequence you would need

$$\Delta \approx 80 \times 0.8 = \mathbf{64\ dB}$$

which is more dynamic range than the cell has. The call drops.

This is the floor rising above the ceiling

When worst-case decay demands a larger Δ than the fluctuations permit, no threshold works and the fix must be *structural*: umbrella cells to keep fast users off the microcell layer, or soft handoff to remove the deadline altogether.

Δ Too Large: the Bus

Back to the macrocell, $P_{\text{min}} = -100$ dBm throughout. Car at **km 1.7**, where $A = -79.9$ and $B = -84.5$ dBm. A bus pulls alongside and shadows the mobile by **11 dB** for 3 s, so A now reads $-$**90.9** dBm. Same event, two choices of Δ:

$\Delta = 8$: $P_{\text{HO}} = -100 + 8 = -92$

-90.9 is still *above* -92.

The threshold is never crossed.

Zero handoffs. The bus passes and nothing at all happened.

$\Delta = 15$: $P_{\text{HO}} = -100 + 15 = -85$

-90.9 is well *below* -85, and B now beats A by 6.4 dB $> H$. It holds 3 s $>$ TTT.

Handoff to B fires. The bus passes, A recovers to -79.9 and beats B by 4.6 dB $> H$, so it **hands back**.

Two handoffs, fifty metres, and how they are prevented

Nothing about the geometry changed; only Δ did. Three filters exist to stop exactly this: **averaging** the samples; **time-to-trigger** (TTT), which demands the condition persist, so a 1.5 s dip would have been ignored where this 3 s one was not; and **hysteresis** H, which is the *relative* margin, not the absolute one. Δ asks how weak the server may get, H how much better the target must be: Δ stops you handing off too late, H stops you handing off back and forth.

Dwell Time

Dwell time is the time over which a user remains within one cell.

Why the statistics matter

Handover algorithms are designed against the *distribution* of dwell time, not a single average. A rare short dwell is what drops calls.

What it varies with

Dwell-time statistics vary greatly with the **speed of the user** and the **type of radio coverage**. A pedestrian in a macrocell and a car in a microcell are different design problems.

Connect it to Δ

Dwell time is the budget; Δ is the spending plan. If typical dwell time falls (smaller cells, faster users), the same Δ that once worked now produces handovers that cannot complete.

Who Decides? Handover Indicators

Network-monitored

Each BS constantly monitors the signal strength of all its reverse voice channels to gauge where each mobile is relative to the BS. This is forwarded to the **MSC**, which decides.

Mobile-assisted (MAHO)

The mobile station measures received power from *surrounding* base stations and continually reports the measurements to its serving BS.

Why MAHO is the better idea

Only the mobile is standing where the call actually is. A base station can measure how well it hears the mobile, but it cannot measure how well the mobile hears a *neighbor*. Pushing measurement to the handset gives the network the one number it genuinely cannot obtain itself.

Prioritizing Handover

The asymmetry that drives everything here

A **dropped call** is considered a more serious event than a **blocked call**. A user denied a new call retries; a user cut off mid-sentence experiences a failure. So channel assignment must give priority to handover requests.

Guard channels

A fraction of the channels in a cell is **reserved only for handover requests**.

Cost: those channels sit idle for new calls, so total carried traffic falls. Dynamic allocation can soften this.

Queuing handover requests

Handover requests are **queued** rather than refused outright, decreasing the probability of forced termination.

Why it works: the margin Δ creates a span where handover is needed but not yet urgent, leaving room to wait, and unlike guard channels, queuing costs no capacity.

Practical Handover: the Speed Problem

The problem

High-speed and low-speed users have vastly different dwell times. High-speed users generate a high number of handover requests, causing interference and traffic-management problems.

The umbrella cell solution

High-speed users are served by large **macro** cells, while low-speed users are handled by small **micro** cells.

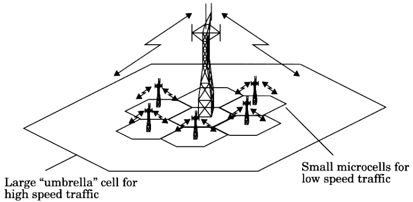


Figure 3.4 The umbrella cell approach.

Rappaport 2e Fig. 3.4: the umbrella cell approach.

The clean idea underneath

Match cell size to user speed so that dwell time is roughly constant across the population. The handover rate, not the cell size, is what the network is really trying to hold steady.

Hard and Soft Handover

Hard handover: break before make

The old channel connection is broken *before* the newly allocated channel connection is set up. This can obviously cause call dropping.

Soft handover: make before break

The new channel connection is established *before* the old one is released. Realized in CDMA, where BS diversity is also used to improve the boundary condition.

Why CDMA can do this and FDMA cannot

In CDMA every cell can use the same frequency, so "listening to two base stations" costs only a second correlator. Under FDMA/TDMA the two links are on different carriers, and the handset has one radio.

Co-Channel Interference and Capacity

The definition

Several cells in a coverage area use the same set of frequencies; these are co-channel cells, and the interference between their signals is **co-channel interference**.

The simplifying assumptions

If all cells are approximately the same size and the path-loss exponent is the same throughout, then the transmit power of each BS is almost equal.

The result worth remembering

Under those assumptions, the worst-case signal-to-co-channel-interference ratio is **independent of transmitted power**. It becomes a function only of the cell radius R and the distance D' to the nearest co-channel cell. Turning every transmitter up does not help: it raises signal and interference together.

Building the SIR Expression

Received power at distance d from the transmitter: $P_r(d) \propto d^{-n}$

Weakest useful signal is at the cell boundary: $P_r(R)$. Interference from a co-channel cell: $P_r(D')$.

The approximation

D' is normally approximated by the base-station separation D between the two cells, unless higher accuracy is required.

General forward link

$$\mathrm{SIR} = \frac{P_r(R)}{\sum_{k=1}^{i_0} P_r(D_k)}$$

D_k: distance of the kth interfering cell from the mobile; i_0: number of co-channel cells.

Why we evaluate at the boundary

The design must work for the *worst* user in the cell, and that user sits at the edge: farthest from the wanted BS and closest to the interferers. Everything better than the boundary case is a bonus.

The First-Order SIR Formula

First tier only, so $i_0 = 6$, and assume $D_k \approx D$ for all k:

$$\mathrm{SIR} \approx \frac{R^{-n}}{6\,D^{-n}} = \frac{(D/R)^n}{6} = \frac{\left(\sqrt{3N}\right)^n}{6} = \frac{(3N)^{n/2}}{6}$$

Worked value

$N = 7,\ n = 4$:

$\mathrm{SIR} = 21^2/6 = 73.5$

$10\log_{10} 73.5 \approx 18.7$ dB

Read the chain

Geometry (N) sets Q, Q sets SIR. This is the single thread connecting every slide so far to every capacity decision that follows.

Outage Probability and SIR Targets

Outage probability

The probability that a mobile station does *not* receive a usable signal.

Required SIR

18 dB for **AMPS**

12 dB for **GSM**

Why GSM tolerates six dB less

GSM is digital and channel-coded, so it survives a worse ratio than analogue AMPS. That lower requirement converts directly into a smaller N, and therefore more channels per cell, as the next slide shows.

Inverting the Model: From SIR Target to Minimum N

With $n = 4$ and $i_0 = 6$: $\quad \mathrm{SIR} \approx \dfrac{(3N)^2}{6} = 1.5\,N^2 \qquad \Longrightarrow \qquad N \geq \sqrt{\mathrm{SIR}_{\text{target}}/1.5}$

AMPS: 18 dB

$10^{1.8} = 63.1; \quad N \geq \sqrt{42.1} = 6.49$

Next allowed cluster size: $\mathbf{N = 7}$

GSM: 12 dB

$10^{1.2} = 15.85; \quad N \geq \sqrt{10.6} = 3.25$

Next allowed cluster size: $\mathbf{N = 4}$

"Next allowed", not "round up"

$6.49 \to 7$ because 7 is the next value $i^2 + ij + j^2$ can produce; $3.25 \to 4$ because $N = 3$ falls short. Here the integer constraint finally bites: it can force a coarser reuse pattern, and so fewer channels per cell, than the arithmetic alone suggests.

What the First-Order Model Leaves Out

Distance approximation

The approximation $D_k \approx D$ is applied from the **second tier onwards** as well. More accurate SIR is obtained by computing the actual distances.

Propagation effects

This computation is based on **path loss only**. Accurate modelling also needs shadowing and fast fading.

Say this out loud in class

Everything computed so far is a *planning estimate*, not a prediction. It tells you which cluster sizes are plausible, then measurement and simulation decide. Treating 18.7 dB as a guaranteed number is the most common misuse of this formula.

Coverage Under Shadowing

At any distance d, measured path loss is random and **log-normally** distributed about its mean:

$$P_r(d)_{\mathrm{dB}} = \overline{P_r(d)}_{\mathrm{dB}} + X_\sigma$$

X_σ: zero-mean Gaussian in dB, standard deviation σ (also in dB).

Boundary coverage

The proportion of locations *at* distance R where signal exceeds a threshold γ (the receiver sensitivity). Expressed with the standard normal Q-function.

Area coverage

The proportion of locations *within* radius R receiving a signal above γ. Obtained by integrating the resulting cdf over the cell area.

Why area coverage always beats boundary coverage

The boundary is the worst ring in the cell; every interior point is closer to the BS. Averaging over the disc therefore lifts the number, which is why 75% at the edge can mean 94% overall.

Coverage: Worked Values and the 90% Rule

Two examples, $\sigma = 8$ dB

$n = 4$, boundary coverage 75%
$\Rightarrow$ **area coverage 94%**

$n = 2$, boundary coverage 75%
$\Rightarrow$ **area coverage 91%**

The operator's criterion

Typically the **"90% rule"**: 90% of a given geographical area must be covered for 90% of the time.

Why larger n helps here

A steeper path-loss exponent makes the mean signal fall off faster with distance, so shadowing is a smaller share of the variation *relative* to the deterministic trend. The graphs are read against σ/n for exactly that reason.

Sizing a Cellular System

The basic design consideration

Sizing the system. It has **two** components, and they are answered by different tools.

Coverage area

Can a user in this location get a usable signal at all? Answered by path loss, shadowing, and the coverage analysis just completed.

Traffic handling capability

Once the signal exists, can the system carry the offered load? Answered by trunking theory, next.

Order of operations

After the system is sized, channels are assigned to cells using the assignment schemes. Sizing first, assignment second: you cannot allocate what you have not dimensioned.

Trunking Vocabulary

Term	Definition
Trunking	Exploiting the statistics of user calling behaviour so that a large number of users can share a limited number of channels.
Grade of service (GoS)	A measure of a user's ability to access a trunked system *during the busiest hour*; typically the likelihood a call is blocked.
Trunking theory	Used to determine the number of channels required to serve a certain offered traffic at a specified GoS.
Call holding time H	The average duration of a call. Note this H is unrelated to the handoff hysteresis H of the mobility section.
Request rate λ	Average number of call requests per unit time.

Note "busiest hour"

GoS is deliberately defined at the worst hour of the day, not the average. A network dimensioned on daily averages fails exactly when people most want it.

Traffic Intensity and the Erlang

Traffic intensity is measured in **Erlangs**, defined as call-minutes per minute:

$A = \lambda \cdot H$

Worked example

There are 3000 calls per hour in a cell, each lasting on average 1.76 minutes.

$A = (3000/60) \times 1.76 = 50 \times 1.76 = \mathbf{88}$ Erlangs

What one Erlang means

One Erlang is one channel kept busy continuously. So 88 Erlangs of offered traffic needs *more* than 88 channels to serve at any decent GoS, because arrivals are random and will sometimes bunch. How much more is exactly what Erlang B answers.

When Traffic Exceeds Capacity

Two strategies for blocking

If offered traffic exceeds the maximum carried traffic, blocking occurs. Systems adopt one of:

- **Blocked calls cleared**: the request is refused and the user retries
- **Blocked calls delayed**: the request waits in a queue

Trunking efficiency

The carried traffic intensity in Erlangs *per channel*, a value between zero and one. It is a function of the number of channels per cell and the GoS parameters.

Call arrivals

It is widely accepted that calls have a **Poisson** arrival process.

The pooling effect, and why it matters later

Trunking efficiency *rises* with pool size: 100 channels in one pool serve far more than 10$\times$ what 10 channels serve. This single fact is the hidden cost of sectoring, which chops one pool into three or six.

Reference Figure: Erlang B

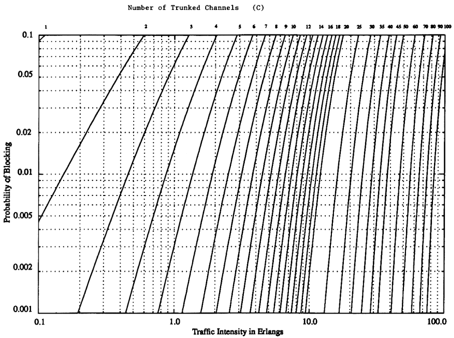


Figure 3.6 The Erlang B chart showing the probability of blocking as functions of the number of channels and traffic intensity in Erlangs.

Rappaport 2e Fig. 3.6: Erlang B blocking probability against offered traffic, one curve per number of trunked channels.

Channel Assignment: Fixed

Fixed Channel Allocation (FCA)

Channels are divided into sets, and a set is **permanently** allocated to each cell. The same set must be assigned only to cells separated by a distance sufficient to control CCI. A call in a cell can be served only by that cell's unused channels; otherwise it is blocked.

Verdict

Easiest to implement, least flexible.

The borrowing modification

A cell that has used all its nominal channels (the **acceptor**) can borrow a free channel from a neighbor (the **donor**).

- Strategies: borrow from the adjacent cell with most free channels, or take the first free channel found
- The borrowed channel must not interfere with existing calls
- It is returned once free

Why borrowing is harder than it sounds

Lending a channel does not just help the acceptor; it silences that channel across the donor's *own* reuse pattern. One local favour propagates as a global constraint.

Channel Assignment: Dynamic and Flexible

Dynamic (DCA)

Voice channels are not allocated permanently. All channels sit in a **central pool** and are assigned as calls arrive.

Each request makes the serving BS ask the MSC, which allocates using the likelihood of future blocking, the reuse distance of the channel, and other cost functions.

Centralized vs. distributed

Centralized DCA uses a single controller for all cells; distributed DCA scatters controllers across the network.

Flexible / hybrid

Split the channels into two groups: one **fixed** per cell, the other a **central pool** shared by all. Mixes the advantages of FCA and DCA; available schemes are *scheduled* and *predictive*.

The cost of DCA

Every one of those cost functions must be evaluated per call, network-wide, in real time. The gain in adaptivity is paid for directly in control complexity and signalling.

Assignment must serve both: new calls *and* handovers, maximizing capacity while minimizing blocking probability.

System Expansion: Two Routes

As demand grows the channels assigned to a cell become insufficient, so more channels must be made available **per unit area**. Note carefully: that is a *density* requirement, not a spectrum requirement.

Route 1: smaller cells

Divide each initial cell area into a number of smaller cells. This is **cell splitting**.

Route 2: more channels per cell

Use a **smaller reuse factor** (smaller N). But to reduce N you must first reduce co-channel interference, which is done by **antenna sectorization**.

Note the dependency

Route 2 is not free either: sectoring must come *first* to buy the SIR headroom that a smaller N then spends. Every capacity gain in this topic is a payment out of the interference budget.

Cell Splitting

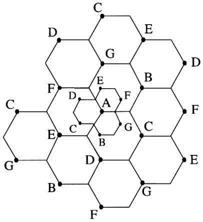


Figure 3.8 Illustration of cell splitting.

Rappaport 2e Fig. 3.8: cell splitting.

What happens

Splitting increases the number of base stations to increase capacity, with a corresponding reduction in **antenna height** and **transmit power**.

Why it is attractive

It accommodates **modular growth**: capacity rises by re-scaling the cellular geometry *without changing the frequency plan*. More cells per unit area means more traffic capacity.

The elegant part

The reuse pattern is scale-free: shrink every R by the same factor and $Q = D/R$ is untouched, so the SIR analysis carries over. That is why splitting needs no new frequency plan.

Power Scaling and the Limits of Splitting

For the new smaller cells, transmit power must be reduced. With $n = 4$, halving the cell radius requires reducing transmit power by a factor of $2^4 = 16$.

Why 2^n: answering the source's "why?"

Received power at the edge goes as $P_t R^{-n}$. To hold edge power fixed while $R \to R/2$:

$$P_t'(R/2)^{-n} = P_t R^{-n} \;\Rightarrow\; P_t' = P_t/2^n$$

Practical limits

In theory splitting repeats indefinitely. In practice it is limited by:

- Cost of base stations
- Handover load, fast vs. slow traffic
- Not all cells split at once, so co-channel interference arises between split and unsplit regions
- Innovative channel assignment is then required

Mixed cell sizes are the real difficulty: during migration the network violates the equal-R, equal-power assumption the SIR analysis rested on.

Sectorization

The aim

Keep the cell radius but **decrease the D/R ratio**. To do that we must reduce the *relative* interference without increasing transmit power.

The mechanism

Sectorization relies on antenna placement and **directivity**. Beams are kept within either a 120° or a 60° sector.

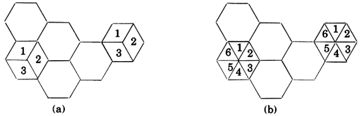


Figure 3.10 (a) 120° sectoring; (b) 60° sectoring.

Rappaport 2e Fig. 3.10: 120° and 60° sectoring.

Read the aim carefully

Lowering D/R means a *tighter* reuse pattern, i.e. a smaller N, i.e. more channels per cell. Sectoring's purpose is not better SIR for its own sake; it is SIR headroom converted into capacity.

Reference Figure: Sectoring Reduces the Interferer Count

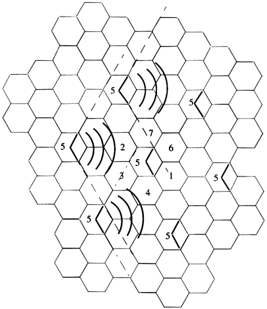


Figure 3.11 Illustration of how 120° sectoring reduces interference from co-channel cells. Out of the 6 co-channel cells in the first tier, only two of them interfere with the center cell. If omnidirectional antennas were used at each base station, all six co-channel cells would interfere with the center cell.

Rappaport 2e Fig. 3.11: with 120° sectoring, only two of the six first-tier co-channel cells interfere with the center cell. With omnidirectional antennas all six would.

Sectorization: Counting the Interferers

First-tier interferers i_0

Omnidirectional	6
Three 120° sectors	**2**
Six 60° sectors	**1**

The cost

Each sector may use only 1/3 or 1/6 of the available channels. So trunking efficiency **decreases**, and the number of required antennas increases.

So how is capacity actually gained?

Not from the sectors themselves: they only fragment the pool. The gain comes from *spending* the SIR improvement: with i_0 cut from 6 to 2, the same SIR target is met at a much smaller N, and $k = S/N$ rises enough to repay the lost trunking efficiency.

Micro Cells

Purpose

Micro cells are introduced to alleviate capacity problems caused by **hotspots**, localized demand peaks that do not justify re-planning the whole area.

Reuse factor

By clever channel assignment, the reuse factor is left **unchanged**.

The recurring difficulty

As with cell splitting, interference problems arise when macro and micro cells must **co-exist**. Two tiers with different sizes and powers break the uniform-geometry assumption behind the reuse plan.

Where this leads

This two-tier structure is the direct ancestor of the modern HetNet, treated in Part 1 of this topic.

Summary

- The cellular concept is divide, separate, reuse: an architecture, not a new radio technology.
- A regular hexagon is assumed for *bookkeeping*; it tiles the plane and covers the most area per site.
- Hexagonal co-ordinates make every cell center an integer pair, and the cosine rule gives $D = \sqrt{i^2 + ij + j^2} \cdot \sqrt{3}R$.
- $N = i^2 + ij + j^2$ is both the reuse-distance factor and the cluster count; $Q = D/R = \sqrt{3N}$.
- Handover needs a margin Δ: too small drops calls, too large floods the MSC. Dwell time sets the budget.
- Dropped calls outrank blocked calls, so handover gets guard channels and queuing.
- First-order SIR is $(3N)^{n/2}/i_0$; inverting it converts an SIR target into a minimum legal N.
- Coverage is separate from capacity: boundary coverage under shadowing is worse than area coverage.
- Trunking converts offered traffic $A = \lambda H$ into a channel count at a target GoS.
- Splitting adds cells; sectoring cuts i_0 from 6 to 2 or 1, buying a smaller N at the price of trunking efficiency.

References

1. `cell-concept.ppt`, "Cellular Systems: Cellular Concepts", 58 slides. Primary source for this deck, followed in its own order.
2. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed., Ch. 3 (pp. 73–136) for cellular geometry, interference, handoff, trunking, splitting and sectoring; Ch. 4 (pp. 138–147) for log-normal shadowing and area coverage.

Reference images in this deck were extracted from the Yale-hosted Pearson/Rappaport Chapter 3 slides, the same figure set used by the source deck: `https://cs.yale.edu/homes/yry/readings/wireless/rappaport_slides/wirelessSlidesCh03.ppt`

Appendix: First-Exposure Terms I

Term	Meaning in this deck
Cell footprint	The actual radio coverage contour of a cell, as opposed to the idealized hexagon
Cluster size N	Number of cells that collectively use the complete channel set once
Channels per cell k	Channels allocated to one cell; $S = kN$
Co-channel cell	A cell using the same frequency set as the target cell
Interference tier	A set of co-channel cells equidistant from the reference cell
First tier	The nearest interference tier; always exactly six cells
Shift parameters (i, j)	Integer steps along the two 60° hexagonal axes
Reuse ratio Q	$D/R = \sqrt{3N}$; co-channel separation relative to cell size

Appendix: First-Exposure Terms II

Term	Meaning in this deck
P_{min}	Minimum usable signal for acceptable voice quality
P_{HO}	Slightly stronger level at which handover is initiated
Δ	Handoff margin, $P_{\text{HO}} - P_{\text{min}}$; absolute, one cell
Hysteresis H	Margin by which the target must beat the serving cell; relative, two cells. Not the trunking holding time H
Time-to-trigger (TTT)	How long a triggering condition must persist before handover starts
dBm / dB	Absolute power referenced to 1 mW / a dimensionless ratio
Dwell time	Time a user remains within one cell
BSC	Base station controller; owns the radio resources of many cells
MAHO	Mobile assisted handover: the handset measures neighbors and reports them
MSC	Mobile switching centre; makes handover decisions from reported measurements
Guard channel	Channel reserved exclusively for handover requests
Hard / soft handover	Break-before-make / make-before-break

Appendix: First-Exposure Terms III

Term	Meaning in this deck
Path-loss exponent n	Rate at which received power decays with distance, $P_r \propto d^{-n}$
i_0	Number of dominant co-channel interferers; 6 omni, 2 for 120°, 1 for 60°
Outage probability	Probability a mobile does not receive a usable signal
Log-normal shadowing	Random path-loss variation, Gaussian when expressed in dB
Boundary coverage	Fraction of locations *at* radius R exceeding the threshold
Area coverage	Fraction of locations *within* radius R exceeding the threshold
90% rule	Operator criterion: 90% of area covered for 90% of the time

Appendix: First-Exposure Terms IV

Term	Meaning in this deck
Trunking	Sharing a limited channel pool among many users with random demand
Grade of service	Blocking quality target, measured in the busiest hour
Erlang	Traffic unit; one channel occupied continuously. $A = \lambda H$
Holding time H	Average call duration
Request rate λ	Average call requests per unit time
Trunking efficiency	Carried Erlangs per channel, between 0 and 1
FCA / DCA	Fixed / dynamic channel allocation
Acceptor, donor	Cell borrowing a channel, and the neighbor lending it
Cell splitting	Subdividing a cell into smaller cells; power scales as 2^{-n} per halving
Sectorization	Dividing a cell into 120° or 60° directional sectors

Appendix: Source Mapping

Deck section	Source slides	Cross-reference
Cellular concept, design problem, cell footprint, hexagon choice	`cell-concept.ppt` 1–5	Rappaport Ch. 3 pp. 73–86; Fig. 3.1
Frequency reuse $S = kN$, terminology, interference tiers	6–7	Rappaport Ch. 3 pp. 73–90
Hexagonal co-ordinates, cosine rule, $N = i^2 + ij + j^2$, $N = 19$ example, Q	8–13	Rappaport Ch. 3 pp. 86–104; Fig. 3.2, Table 3.1
Handover definition, threshold Δ, dwell time, MAHO, prioritization	14–19	Rappaport Ch. 3 pp. 112–123; Fig. 3.3
Umbrella cells, hard and soft handover	20–22	Rappaport Ch. 3 pp. 112–123; Fig. 3.4
CCI, received-power model, forward-link SIR, first-tier formula	23–27	Rappaport Ch. 3 pp. 90–104
Outage probability, SIR targets, minimum N, model caveats	28–31	Rappaport Ch. 3 pp. 90–104
Shadowing, boundary and area coverage, 90% rule	32–37	Rappaport Ch. 4 pp. 138–147; Fig. 4.18
Sizing, trunking vocabulary, Erlangs, blocking strategies	38–41	Rappaport Ch. 3 pp. 123–131; Fig. 3.6
FCA with borrowing, DCA, flexible assignment	42–45	Rappaport Ch. 3 pp. 123–131
Expansion routes, cell splitting, power scaling, limits	46–50	Rappaport Ch. 3 pp. 131–136; Figs. 3.8–3.9
Sectorization, interferer counts, micro cells	51–58	Rappaport Ch. 3 pp. 131–136; Figs. 3.10–3.11